Yanis Bérard

Student at the École Normale Supérieure de Rennes – Mathematics

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Education

ENS Rennes *2025 – Present*
Student (*élève normalien*), Pure Mathematics track (*Magistère de Mathématiques*).

Université de Rennes *2026 – Present*
Master's degree in Mathematics, first year (M1).

Université de Rennes *2025 – 2026*
Bachelor's degree in Mathematics, third year (Licence 3).

Lycée du Parc, Lyon *2024 – 2025*
CPGE MP*1 (two-year undergraduate preparatory classes for the Grandes Écoles).

Lycée La Martinière Monplaisir, Lyon *2022 – 2024*
CPGE MPSI → MP*.

Lycée Honoré d'Urfé, Saint-Étienne *2022*
Baccalauréat, Mathematics and Physics majors, with highest honours (*mention Très Bien*).

Research Experience

Research internship, Institut Camille Jordan, Lyon *May – June 2026*
Supervised by Jean-Yves Welschinger. Report: *Local Holomorphy and Global Algebraicity: The Road to GAGA* (compact Riemann surfaces, covering theory and the Riemann–Hurwitz formula, sheaf cohomology, Riemann–Roch theorem, Serre duality, projective embedding, Chow's theorem, GAGA for curves).

GATE Laboratory, Saint-Étienne *2023*
Study of voting theory and numerical simulations around Arrow's impossibility theorem.

Academic Projects

- **Expository paper (2026):** *An elementary proof of Bézout's theorem on algebraic curves.* Study of intersections of plane projective curves, local intersection multiplicities, explicit examples and geometric interpretation. Three-hour presentation to fellow students.
- **TIPE (supervised research project):** Lattices of finite groups and applications; random walks on a group and the Poisson boundary.

Teaching, Outreach & Other Experience

Private tutor in Mathematics and Physics *2021 – Present*
Regular tutoring of students from middle school to preparatory classes (CPGE), including preparation for the competitive entrance exams of the Grandes Écoles.

Outreach: Organisation of the *Rencontres des Jeunes Mathématiciennes et Informaticiennes* (outreach event for young women in mathematics and computer science), design of a workshop.

Mechanical worker – *NTD France*; **Market gardening** – *Ferme Le Bénitier aux Oiseaux*

Skills

- **Languages:** French (native), English (C1), Portuguese (B2), Spanish (B1), Italian (B1).
- **Computer skills:** LaTeX, Python, OCaml.

Interests

Climbing (Loire junior champion), skiing, running, cycling, chess.